



# A toolkit for the analysis and visualization of free volume in materials

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## ABSTRACT

A suite of tools is presented which enable analysis of free volume in terms of accepted standard metrics. The tools are extensible through the use of standard UNIX tools to be useful for output of many standard simulation packages. The tools also include utilities for rapid development of visual output not available in other packages. The tools are also extensible and modifiable for other types of spatial data.

## Categories and Subject Descriptors

H.2.8[Database Applications] Data mining;

## General Terms

Algorithms, Measurement, Design, Standardization, Theory

## Keywords

Free volume, shape, morphometrics, virtual measurement, molecular dynamics, monte carlo, visualization, materials, separations

## 1. INTRODUCTION

Free volume properties characterize transport behavior in molecular systems. In particular, the shape and connectivity of free volume determine permselectivity and diffusivity of penetrating diffusant species.<sup>1</sup>

Classical and quantum molecular dynamics packages focus on atoms and groups of atoms. By contrast, the tools presented implement well-known algorithms for the characterization of free volume, including energetic cavity sizing, insertion factor calculation, and rattle-in-place measures. The tools are UNIX-based and leverage the standard ‘UNIX plumbing’ of sed/awk/grep and pipes for the development of efficient workflows. Workflows can be created for nearly any type of molecular data, and have been modified to work with re-sampled microscope data as well. There are additional tools for converting to Povray<sup>2</sup> Scene Definition Language (SDL) as well as to and from TIFF formats.

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Tools such as these for gathering shape data will likely play an important role in the analyses used e.g. for protein docking and fit in well with advances in shape metrology which borrow from computer vision.<sup>3</sup>

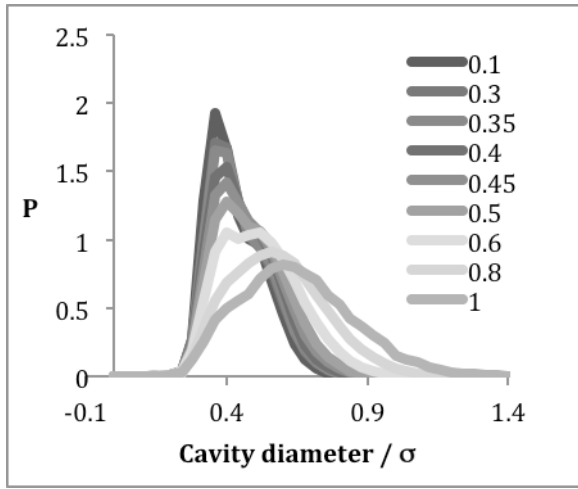
The popular VMD<sup>4</sup> package works in conjunction with NAMD to provide publication grade visuals from classical molecular dynamics simulations. Other popular viewers include the lightweight Jmol<sup>5</sup> which reads PDB, CIF and many other standard formats and V3D<sup>6</sup> which enables viewing of 3D microscope data. To the best of the author’s knowledge, this is the first package to be developed *explicitly* for the calculation and visualization of negative space in polymers.

Sample results are presented for Simmons’ coarse-grained model glass former<sup>7</sup> simulated using the techniques of classical molecular dynamics and for a classically simulated atomistic sample of poly-dimethylsiloxane (PDMS) at 298K.

## 2. ANALYTICS OF FREE VOLUME

### 2.1 Cavity Sizing

*ddx* is a C-language implementation of Veld’s Cavity Energetic Sizing Algorithm<sup>8</sup> (CESA). It is a Monte Carlo algorithm which searches out the local minima of insertion energy for a test particle, then resizes the test particle to fill the void. In the algorithm, a test point is selected at random within the sample. A Lennard-Jones ghost atom of zero diameter (Berthelot combining rule) is inserted at the test point, then allowed to travel via the steepest descent to the lowest energy position available to it. Upon reaching this position, the particle is scaled to such a point where its repulsive and attractive interactions with the sample material are balanced. The object thus obtained is recorded as a ‘cavity’. The code allows switching on or off the requirement that the original insertion point be contained within the scaled test particle, as this is the mechanism for enforcing volume sampling of the cavity in the original algorithm. The output is a tabulated list of cavities thus sized. The utility *cav2pov* converts the list to SDL.



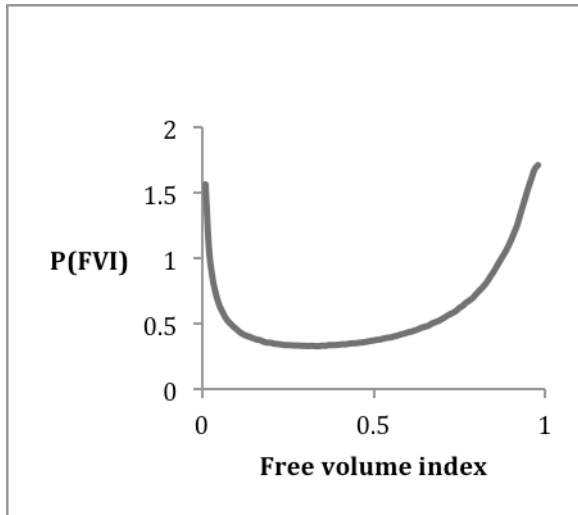
**Figure 1.** The volume-weighted cavity size distributions for the simulated samples of Simmons' glass over a range of reduced temperatures. The glass, crossover, and Arrhenius temperatures for the system are  $T_g = 0.37$ ,  $T_c = 0.42$ , and  $T_A = 0.59$ .

## 2.2 Free Volume Index

*gfg2fvi* is a CUDA-based utility which performs Widom-type insertions<sup>9</sup> of a ghost particle at regularly spaced grid points to determine a free volume index<sup>10</sup>:

$$FVI(x, y, z, t) = e^{-\beta \psi_{ms}}$$

where  $\beta = 1/k_B T$  and  $\psi_{ms}$  represents the interaction of the ghost particle with the system. Additionally, the output of *gfg2fvi* can be piped to the utility *fvi2tiff* which generates a grayscale 3D TIFF of the free volume index, or to the utility *fvi2pov* which generates SDL.

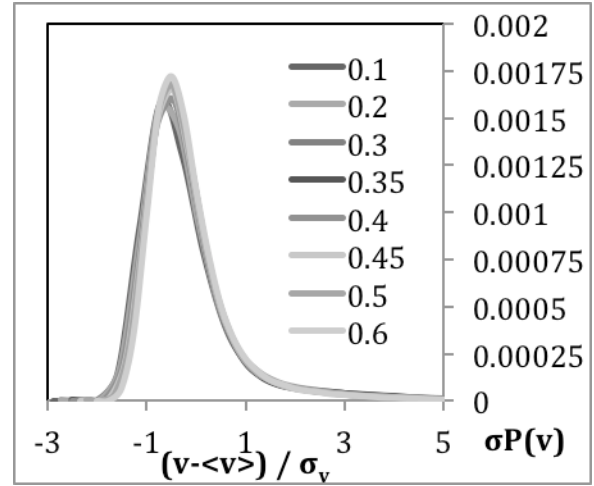


**Figure 2.** The free volume index distribution for a simulated sample of PDMS. Peaks at both extremities indicate clear distinction between occupied and unoccupied space. The index for the sample is mapped to the green channel of a TIFF to generate the image shown in Figure 5.

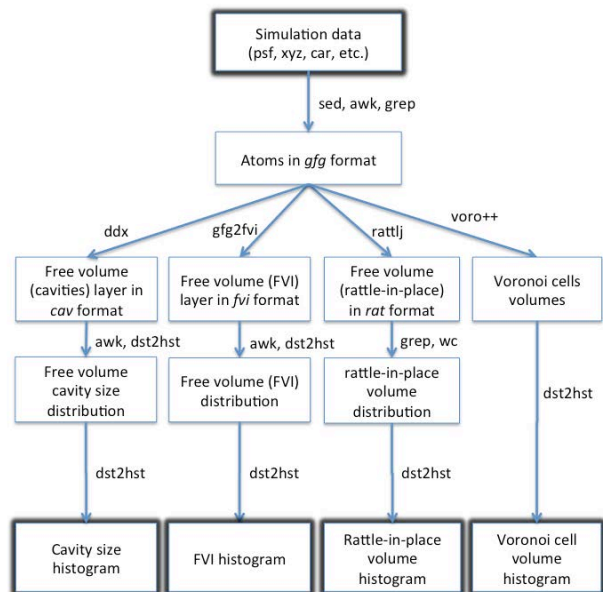
## 2.3 Rattle-in-place

*rattlj* is a C-language code which performs a recursive search of the space connected to each of the centers of a chain of Lennard-Jones atoms. Contiguous spaces for which the re-insertion energy of the removed center is below the energy threshold ( $2kT$ ) are included in the measure. The sets of contiguous spaces thus generated can be regarded as the 'rattle-in-place volumes' for the centers. It is noted that this differs from the original<sup>11</sup> formulation of rattle volumes, which used a geometric definition. The spaces generated can be knocked down to SDL by a series of UNIX pipes through the *fvi2pov* utility.

Figure 3 shows rattle-in-place results adjusted by mean and rescaled by the standard deviation, as done by Starr<sup>12</sup> et. al. It was noted by Starr that the collapse is nearly identical for all temperatures. It is noted here that a clear though faint trend with temperature emerges for the system studied this way. The Starr result was for a system maintained at constant density, whereas this is for a system maintained at constant pressure.



**Figure 3.** The rattle-in-place volumes as determined for a range of temperatures using *rattlj*.

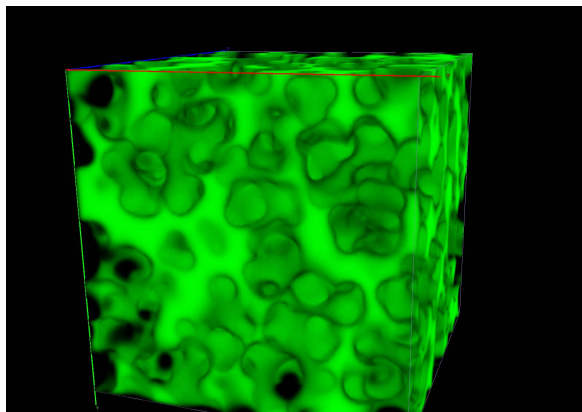


**Figure 4.** Example workflow for generating histogram data for free volume virtual metrics.

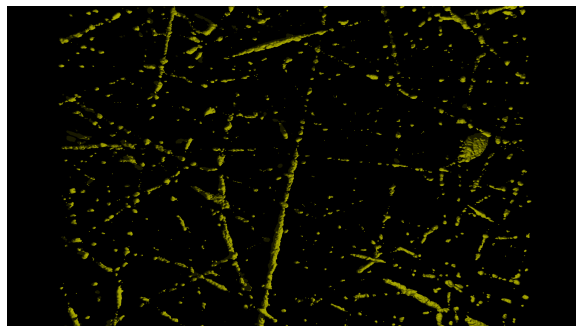
### 3. VISUALIZATION OF FREE VOLUME

#### 3.1 TIFF Interoperability

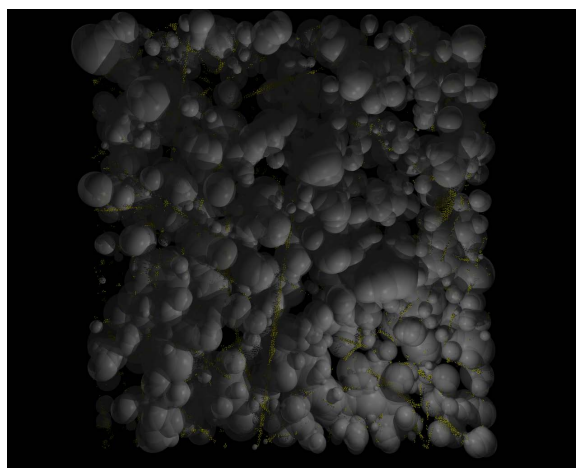
*fvi2tiff* allows for FVI maps to be written to a 3D TIFF, which can then be viewed using a standard viewer as in figure 5. It is possible to use the API to read in TIFF data such as confocal microscope data. Figure 6 shows a resampled image of an electrospun polymer tissue scaffold and the accompanying cavities generated by running *ddx*, providing a virtual metric for the free space in the scaffold.



**Figure 5.** The free volume index of a simulated sample of PDMS. Image is mapped to 3D TIFF format and rendered using V3D



**Figure 6a.** A stack of TIFF images of an electrospun tissue scaffold is resampled and reconstructed as a set of ‘pseudo-atoms’.



**Figure 6b.** The tissue scaffold ‘pseudoatoms’ are analyzed with *ddx* to generate a set of cavities, the sizes of which are characteristic of the free space in the scaffold.

#### 3.2 Managing Image Layers

It is possible to specify transparency levels, ‘phong’, color and a host of other options for each layer of a given image, as shown in this example usage statement for *gfg2pov*:

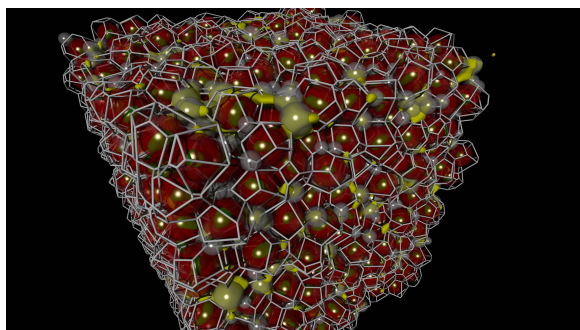
```
usage:      gfg2pov  -transmit [ 0.0 ]
               -phong [ 0.0 ]
               -color [ White ]
               -clip
*** Header parameters and defaults ***
-light_source [ 125.0 125.0 125.0 ]
-light_color [ White ]
-standard_light
-camera [ 40.0 40.0 40.0 ]
-look_at [ 0.0 0.0 0.0 ]
-box [ 10.0 10.0 10.0 ]
-background [ Black ]
```

```

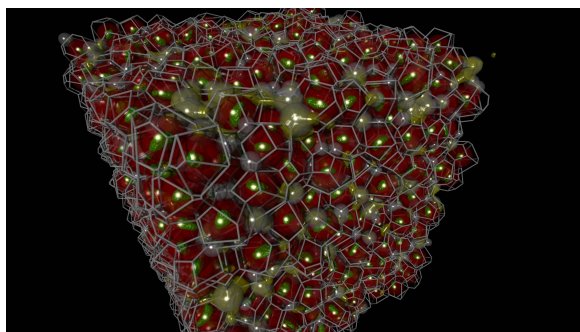
-ambient_light [ 0.0 ]
-right [ 1.0 (for 16/9 aspect ratio) ]
-show_floor
-show_box
-shadowless
-no_header

```

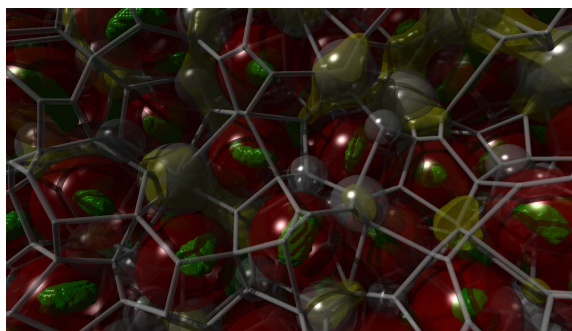
By capturing individual layers, it is possible to generate a header and individual components, then specify which layers to include via a UNIX ‘cat’ statement in a Makefile. Figures 6a through 6e show inclusion of different layers and different command line arguments.



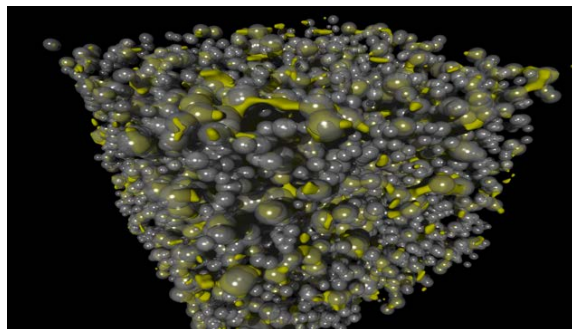
**Figure 7a.** Rendering of Simmons’ glass (red bubbles) with CESA cavities (gray bubbles), free volume index (yellow blob) and rattle-in-place volume (green blob). Also shown is the Voronoi diagram (gray tubules) for the fluid. Notably, this illustrates the mathematical dual relationship of the Veld cavities (gray) which center on the corners of the Voronoi diagram and the atoms (red) which center on the corners of the Delaunay diagram. The voro++ package<sup>13</sup> was used to generate the Voronoi layer.



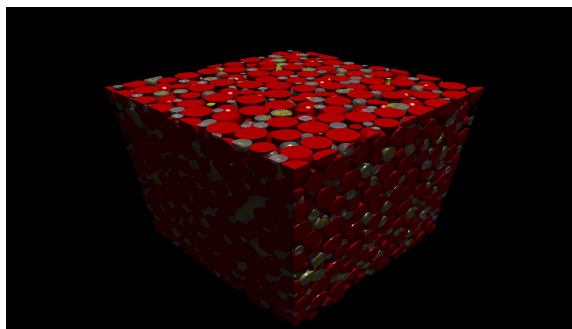
**Figure 7b.** Same image showing transparent layer for FVI.



**Figure 7c.** Same image with different camera placement (close-up).



**Figure 7d.** Only the FVI and cavity layers.



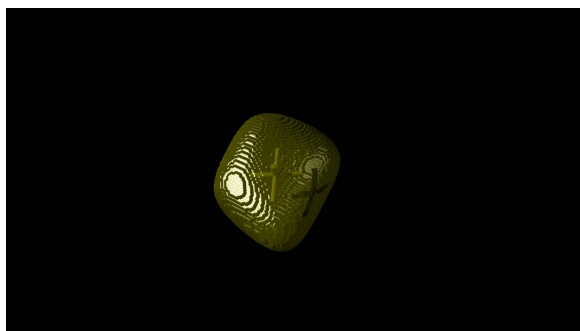
**Figure 7e.** Atoms, cavities, and FVI layer, with clipping at periodic boundary.

It is also possible to use scripts to generate parametric output to be fed to the utilities in order to build frames to generate a high-quality moving image (not shown.)

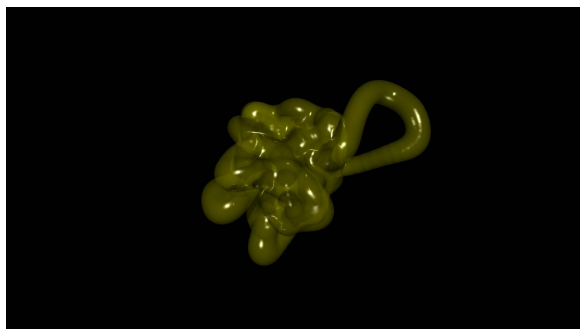
### 3.3 Extensibility

Because the toolkit operates on standard UNIX utilities, the tools are readily extensible for the processing and viewing of other types of data. A couple of examples are provided in figure 8.

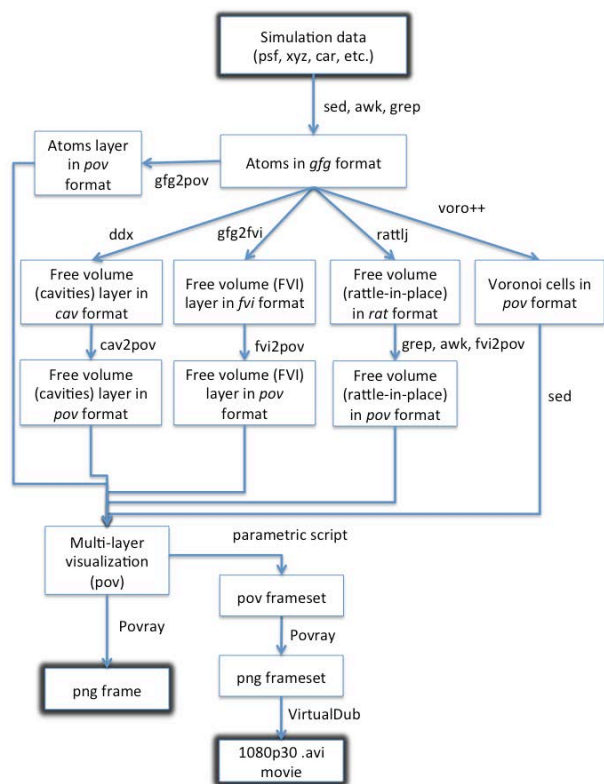




**Figure 8a.** A principal component analysis was performed on the dataset representing an individual rattle-in-place volume using the statistical package R<sup>14</sup>. The result was added as a layer and the image rendered. The orientation of the cross is along principal axes. Lengths of axes represents magnitude of component.



**Figure 8b.** The trajectory for an individual atom, captured for rendering via UNIX 'awk' and *fvi2pov*.



**Figure 9.** A sample workflow for generating single-frame free volume images or HD movies from simulation data.

## 4. CONCLUSIONS

A set of tools for capturing shape information of free volume elements is presented. The methods implement standard free-volume techniques in a way that enables visualization not previously possible. The methods implemented in these codes enable fast work-up of publication quality visuals. These methods are extensible and enable viewing of free volume as well as other spatial data layers.

Please contact the author directly to obtain the software.

## 5. ACKNOWLEDGEMENTS

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